

Fast or Accurate?

Comparing Deep-Learning and Model-Based Wavefront Sensing
from Phase-Diverse PSFs, and Combining Them

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Abstract

Focal-plane wavefront sensing estimates optical aberrations directly from point-spread-function (PSF) images, without a separate wavefront sensor. Two kinds of estimator are widely used: iterative model-based fitting, which is accurate but slow, and neural networks, which are fast but less accurate. We compare both in a controlled simulation with a 16-mode Zernike wavefront, a two-channel (in-focus plus defocused) PSF input, Poisson and read noise, and a Kolmogorov-like modal spectrum. A compact convolutional neural network (CNN) with 1.2 million parameters reduces the mean residual wavefront RMS on validation data from 0.549 rad to 0.077 rad (Strehl ratio 0.724 to 0.993). Removing the defocused channel raises the residual to 0.214 rad, with a per-mode error pattern consistent with phase diversity resolving the sign ambiguity of even-order modes; this ablation used fewer training epochs for the single-channel model (7 versus 15). A gradient-descent physics fit using the same renderer as the data generation reaches 0.024 rad (Strehl 0.999) in 37.9 s for 200 frames, so the CNN is about $53\times$ faster on CPU but about $3\times$ less accurate. Robustness sweeps over photon flux and turbulence strength show where the CNN remains effective. We also propose a hybrid estimator in which the CNN prediction initialises the physics fit; it has not yet been evaluated. All results come from a single simulation run (one random seed) and do not represent performance on a real optical system.

Keywords: wavefront sensing; phase diversity; adaptive optics; Zernike polynomials; convolutional neural networks; point-spread function

1 Introduction

In adaptive optics (AO) systems, wavefront distortions caused by atmospheric turbulence are compensated, usually in closed loop at high frame rates. Conventional systems employ a separate wavefront sensor, typically a Shack–Hartmann sensor. Focal-plane wavefront sensing instead assumes that the wavefront can be determined from images taken at or close to the science focus, which helps minimise non-common-path errors and simplifies the optical train. The PSF depends on the squared modulus of the Fourier transform of the pupil field, so part of the phase information is lost; in general, a single in-focus image cannot determine the sign of even-order aberrations (e.g., defocus or astigmatism).

This ambiguity is overcome by phase diversity, which uses a second image taken with an additional known aberration, here defocus [1, 2]. Classical estimators fit a parametric wavefront model to both images by nonlinear optimisation. These fits are accurate if the forward model is correct, but each iteration requires several evaluations of the FFT-based PSF, which is costly.

More recently, convolutional neural networks (CNNs) have been applied to predict aberration coefficients from PSF images in a single forward pass [3–5]. CNN-based wavefront estimation is therefore well established, and the novelty of this paper does not lie in the existence of a CNN estimator. We instead make three contributions:

- a head-to-head evaluation of the CNN and the physics-based fit on the same simulated scenario, in both accuracy and speed;
- an operating-regime map of the CNN as a function of photon flux and turbulence strength; and
- a proposed hybrid estimator combining the physics fit and the CNN output. The hybrid is described but has *not* been evaluated in this study.

All results are simulation results from a reduced-size training configuration with a single random seed. They should be read as a controlled comparison, not as on-sky or laboratory performance.

2 Related Work

Phase retrieval and phase diversity. Wavefront sensing from intensity measurements by phase retrieval was introduced by Gonsalves and Chidlaw [1], and the use of an additional, deliberately defocused image (phase diversity) in adaptive optics was described by Gonsalves [2]. Iterative phase-retrieval algorithms have been compared extensively [6]. These model-based methods need an explicit forward model of the imaging system.

Learning-based wavefront sensing. Several machine-learning approaches have been used for wavefront sensing from images. Paine and Fienup initialised image-based phase retrieval with a machine-learning estimate [3]. Nishizaki et al. and Swanson et al. estimated wavefront aberrations from intensity images using deep neural networks [4, 5]. The hybrid estimator proposed here is inspired by the same idea of combining a learned estimate with a physics-based refinement.

Turbulence and modal representation. Zernike polynomials are a standard modal basis for turbulence-induced aberrations, and Noll [7] derived their statistics under Kolmogorov turbulence. Our simulator is a modal model rather than a full spatial phase-screen simulation, using a Kolmogorov-like modal variance decay.

Matched-simulator evaluation. Reported accuracy can be misleading when the same numerical model is used both to generate the data and to invert it, a situation known as an inverse crime [8]. This directly affects the interpretation of the physics-fit results in this paper (Sections 7 and 8).

3 Methodology

3.1 Wavefront and imaging model

The pupil-plane phase is represented as a sum of 16 Noll-normalised Zernike modes [7],

$$\varphi(\mathbf{u}) = \sum_{j=1}^{16} c_j Z_j(\mathbf{u}), \quad c_1 = 0, \quad (1)$$

where c_j is the coefficient of mode Z_j in rad RMS and piston ($j = 1$) is set to zero. The standard deviation of the coefficients scales with radial order n as $n^{-11/6}$, giving a Kolmogorov-like modal spectrum with most of the energy in low-order modes. The total RMS wavefront error of each

sample is rescaled to be uniformly distributed between 0.2 and 1.0 rad. This is not a complete spatial Kolmogorov phase screen.

PSFs are computed by Fourier transform on a 64×64 grid of the pupil field. The pupil diameter is 32 px and the resolution element λ/D is 2 px (Nyquist sampling). Two channels are used for each sample,

$$I_k(\mathbf{x}) = |\mathcal{F}\{P(\mathbf{u}) \exp[i(\varphi(\mathbf{u}) + \delta_k Z_4(\mathbf{u}))]\}(\mathbf{x})|^2, \quad k = 1, 2, \quad (2)$$

where $P(\mathbf{u})$ is the pupil function, $\mathcal{F}$ the Fourier transform, and the diversity coefficients are $\delta_1 = 0$ (in-focus channel) and $\delta_2 = 1.0$ rad RMS of defocus (Noll Z_4 ; diversity channel). Poisson photon noise and Gaussian read noise (5 electrons RMS) are applied to each frame. Each image is square-root compressed and normalised by its maximum value to $[0, 1]$, because the dark parts of the images are typically much fainter than the brightest parts.

3.2 CNN estimator

The CNN takes the two-channel PSF stack and regresses the 16 Zernike coefficients. In the CPU (“small”) profile used here, the network has approximately 1.2 million parameters, six 3×3 convolution blocks (three with stride 2), and a 256-unit fully connected head. No global pooling is used, so PSF position (tip/tilt) information is retained. The small model was selected automatically because no GPU was available; a ResNet-18 variant [9] was prepared for GPU use but was not tested. Training uses a mean-squared-error loss and the AdamW optimiser [10], with settings given in Section 4.2.

3.3 Physics-based fit

The physics-based estimator minimises the mismatch between the measured two-channel PSFs and the PSFs predicted by the differentiable forward model of Eq. (2), optimising the 16 coefficients by gradient descent with the Adam optimiser [11] in PyTorch [12]. The objective is the sum of squared differences between observed and rendered square-root-compressed images,

$$L(\mathbf{c}) = \sum_{k=1}^2 \sum_{\mathbf{x}} [\tilde{I}_k(\mathbf{x}) - \hat{I}_k(\mathbf{x}; \mathbf{c})]^2, \quad (3)$$

where $\tilde{I}_k$ is the observed (compressed and normalised) image and $\hat{I}_k(\mathbf{x}; \mathbf{c})$ the rendered image. Optimisation starts from small random initial parameters (Gaussian, $\sigma = 0.05$ rad) and runs 200 Adam steps with a learning rate of 0.05, batched over frames. Importantly, the fit uses the same renderer that generated the data, so the forward model is exact. This matched-simulator condition (an inverse crime [8]) favours the physics fit and is discussed further in Sections 7 and 8.

3.4 Proposed hybrid estimator

In the hybrid estimator, the CNN prediction replaces the random initial guess of the physics fit,

$$\mathbf{c}^{(0)} = f_{\theta}(\tilde{I}_1, \tilde{I}_2), \quad (4)$$

where f_{θ} is the trained CNN, followed by the same 200-iteration fit. The quantities of interest would be the number of iterations needed to reach a given residual compared with the standard random start, and the wall-clock time. The implementation change is small, but the hybrid has *not* been run and no hybrid results are reported; it is listed as future work (Section 8).

4 Experimental Setup

4.1 Data

The CPU profile used 3000 training and 500 validation samples. The photon count of each training frame was sampled log-uniformly between 10^3 and 10^6 , while validation frames used 10^5 photons. Input RMS wavefront error ranged from 0.2 to 1.0 rad for both training and validation. A larger GPU profile (8000 training and 1000 validation samples, ResNet-18, 25 epochs) was defined but not run, so the CNN results below should be interpreted with caution.

4.2 Training configuration

The CNN was trained for 15 epochs with batch size 64 using AdamW (learning rate 10^{-3} , weight decay 10^{-4}), with a 1-epoch warm-up followed by cosine decay to 1% of the initial learning rate, and gradient clipping at 1.0. The checkpoint with the lowest validation loss was kept.

4.3 Evaluation experiments

- **Channel ablation.** A one-channel (in-focus only) CNN was trained and compared with the two-channel CNN. The two-channel model was trained for 15 epochs and the single-channel model for $\max(5, \lfloor 15/2 \rfloor) = 7$ epochs.
- **Photon-flux sweep.** The trained CNN was tested without retraining on 300 new samples at each of 10^3 , 3×10^3 , 10^4 , 10^5 and 10^6 photons/frame.
- **Turbulence sweep.** The trained CNN was evaluated without retraining on 300 fresh samples for each input-RMS range (0.1, 0.3), (0.3, 0.6), (0.6, 1.0) and (1.0, 1.5) rad.
- **CNN vs. physics fit.** Both estimators were applied to the first 200 validation samples, and wall-clock time was measured on CPU.

4.4 Reproducibility and environment

All simulations and training were run in a single CPU session using a custom 1.2-million-parameter CNN implemented in PyTorch and NumPy (seed 42) for 15 epochs, with a training time of 575 s. The quantitative results are reported in Section 6.

5 Evaluation Metrics

Residual wavefront RMS. The residual is the Euclidean norm of the coefficient error excluding piston,

$$\sigma = \sqrt{\sum_{j=2}^{16} (c_j - \hat{c}_j)^2}, \quad (5)$$

where $\hat{c}_j$ are the estimated coefficients. Because the modes are Noll-normalised, σ is in rad RMS. The mean over evaluated samples is reported; this draft does not report the median or 90th percentile.

Strehl ratio. The Strehl ratio S is the ratio of the on-axis intensity of the PSF to that of the ideal (aberration-free) PSF. It is computed from the rendered PSF, so tip/tilt errors reduce it. For small aberrations it is approximately related to the residual by the Maréchal approximation [13],

$$S \approx \exp(-\sigma^2). \quad (6)$$

This is used only as an approximate consistency check, and it holds approximately for the reported values: $\sigma = 0.549$ rad gives $\exp(-\sigma^2) \approx 0.74$ against a measured Strehl of 0.724, and $\sigma = 0.077$ rad gives ≈ 0.994 against 0.993.

Table 1: Main CNN results on validation data (single run, reduced profile).

Metric	Uncorrected	CNN
Residual RMS (rad)	0.549	0.077
Strehl ratio	0.724	0.993
Fixed-range PSNR (dB)	24.57	41.46
Fixed-range SSIM	0.1768	0.9460

Phase-map PSNR and SSIM. Reconstructed phase maps are compared with true maps using a fixed-range peak signal-to-noise ratio,

$$\text{PSNR} = 10 \log_{10} \left(\frac{R^2}{\text{MSE}} \right), \quad (7)$$

together with the structural similarity index (SSIM) [14], where R is the maximum phase-to-phase difference within the set of maps being compared. An earlier version of the notebook normalised each map individually (27.79 dB / 0.9442). This is replaced because per-map normalisation can mask amplitude inaccuracies, and that variant is not used in the results presented here.

Coefficient MSE and timing. During training the validation mean-squared error on the coefficients is reported. Timing is the wall-clock time for 200 frames on CPU.

Validation of the numerical basis. Zernike polynomials are orthogonal on the continuous unit disc but not necessarily on a pixel grid. The Gram matrix of the discretised basis is close to diagonal, with a mean diagonal of 0.976 and a maximum off-diagonal entry of 0.052, so the modes are close to orthonormal and the coefficients remain meaningful.

6 Results

All values in this section come from a single run with seed 42 in the reduced CPU profile.

6.1 CNN training and reconstruction quality

The validation MSE of predicting zero for all modes is 0.02457. The CNN reaches 0.01057 after 1 epoch and 0.00048 (training 0.00037) after 15 epochs, about a 98% improvement over the zero baseline. The mean residual RMS decreases from 0.386 rad at epoch 1 to 0.077 rad at epoch 15, against an uncorrected input RMS of 0.549 rad. The gap between training and validation loss is not large enough to indicate strong overfitting.

With fixed-range metrics, the CNN-corrected phase maps achieve a PSNR of 41.46 dB and an SSIM of 0.9460, whereas the uncorrected baseline phase maps yield 24.57 dB and 0.1768. The mean Strehl ratio improves from 0.724 to 0.993 (Table 1, Fig. 1).

6.2 Phase-diversity ablation

When the defocused channel is removed and only the in-focus PSF is used, the residual rises to 0.214 rad (about $2.8\times$ larger) and the Strehl ratio falls to 0.947 instead of 0.993 (Table 2). The per-mode error chart (Fig. 2) shows that the largest degradation occurs for defocus (Z_4) and astigmatism (Z_5, Z_6). This is consistent with expectation: for even-order modes, a single in-focus PSF is insensitive to the sign of the coefficient, and the single-channel network predicts values close to zero for them. The diversity channel resolves this ambiguity.

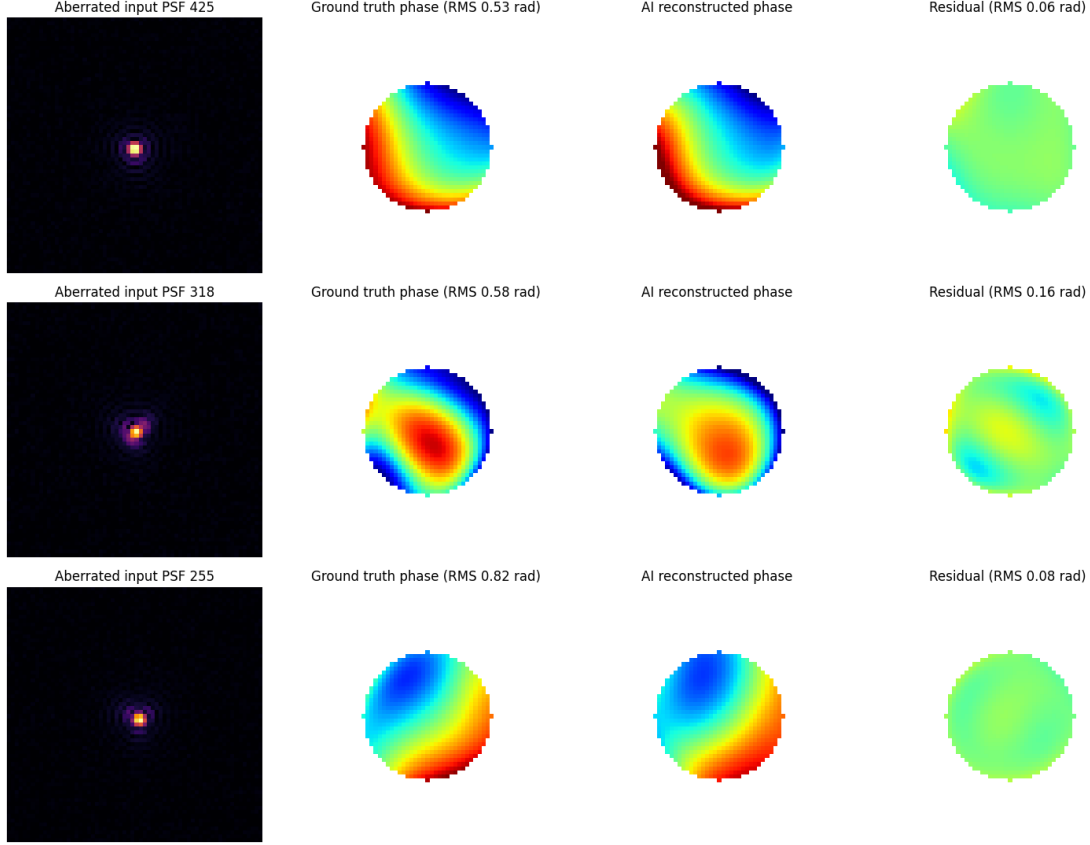


Figure 1: True, uncorrected and CNN-corrected wavefronts for representative validation samples.

Table 2: Channel ablation. The single-channel model was trained for 7 epochs and the two-channel model for 15.

Input	Residual RMS (rad)	Strehl
1 channel (in-focus)	0.214	0.947
2 channels (in-focus + defocus)	0.077	0.993

Limitation of this comparison. The single-channel model was trained for 7 epochs while the two-channel model was trained for 15. The two models therefore differ both in input and in training time, and part of the difference between 0.077 and 0.214 rad is likely attributable to the shorter training of the single-channel model. The ablation should be repeated with equal training time before the size of the effect is attributed solely to the second channel.

6.3 Robustness to photon flux

At 10^3 photons per frame, the CNN reduces the RMS from 0.565 to 0.245 rad (Strehl 0.714 to 0.938). As a conventional but approximate reference, a residual of 0.45 rad corresponds to a Strehl ratio of about 0.8; the 10^3 -photon residual of 0.245 rad is below this value. Only the endpoints of the sweep are tabulated (Table 3); intermediate flux settings are found in `results.json` and Fig. 3.

The training photon counts ranged from 10^3 to 10^6 , so every setting in the sweep lies within the training range. This sweep therefore measures performance under noise conditions the network was trained on, and is not an out-of-distribution or extrapolation test. No retraining

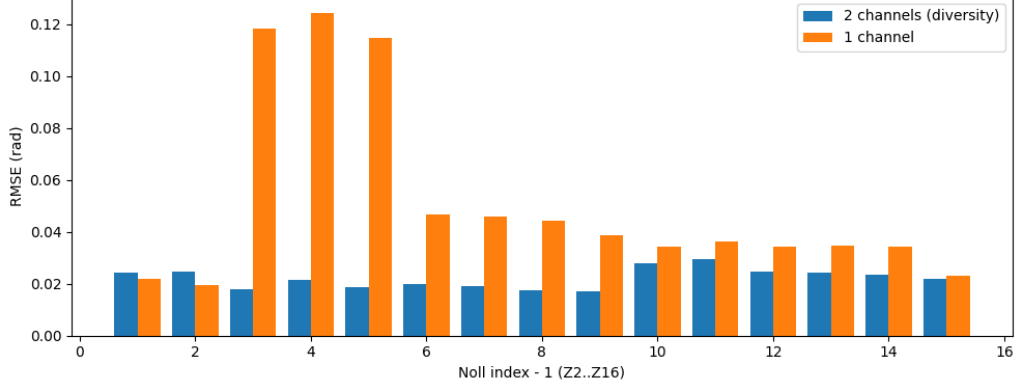


Figure 2: Per-mode coefficient error for the one-channel and two-channel CNNs (unequal training epochs: 7 and 15).

Table 3: CNN robustness (endpoints of each sweep). RMS and Strehl are means over 300 samples.

Condition	Input RMS (rad)	Residual RMS (rad)	Strehl before → after
10^5 photons	0.542	0.078	0.730 → 0.992
10^3 photons	0.565	0.245	0.714 → 0.938
Mild turbulence, range (0.1, 0.3)	0.181	0.033	0.961 → 0.999
Strong turbulence, range (1.0, 1.5)	1.170	0.260	0.356 → 0.914

was done, and 300 new samples were used per setting.

6.4 Robustness to turbulence strength

Under mild turbulence (input RMS range 0.1–0.3 rad) the mean RMS falls from 0.181 to 0.033 rad (Strehl 0.961 to 0.999). Under strong turbulence (range 1.0–1.5 rad) it falls from 1.170 to 0.260 rad (Strehl 0.356 to 0.914). The residual error grows with input strength, but the correction remains considerable even for strongly distorted PSFs.

The training distribution (0.2–1.0 rad) relates to these sweeps as follows. The intermediate ranges (0.3, 0.6) and (0.6, 1.0) lie within it and so test interpolation. The lowest range (0.1, 0.3) overlaps the training range only partially, and the highest range (1.0, 1.5) lies completely outside it. The strong-turbulence endpoint is thus out of distribution, and the mild-turbulence endpoint is partly so. Values for the intermediate ranges are not given here.

6.5 CNN versus physics fit

On 200 frames the physics fit attains a residual of 0.024 rad (Strehl 0.999) in 37.9 s, whereas the CNN attains 0.074 rad (Strehl 0.993) in 0.711 s (Table 4). On this CPU implementation the CNN is therefore about 53 times faster, and the physics fit is about 3 times more accurate in residual RMS. The CNN residual of 0.074 rad differs slightly from the 0.077 rad in Table 1 because it is computed on a different subset (the first 200 validation samples instead of the full validation set).

7 Discussion

Speed versus accuracy. In this study the CNN is less accurate than the physics fit. Its advantage is computational: it processed 200 frames in 0.711 s (about 280 frames per second)

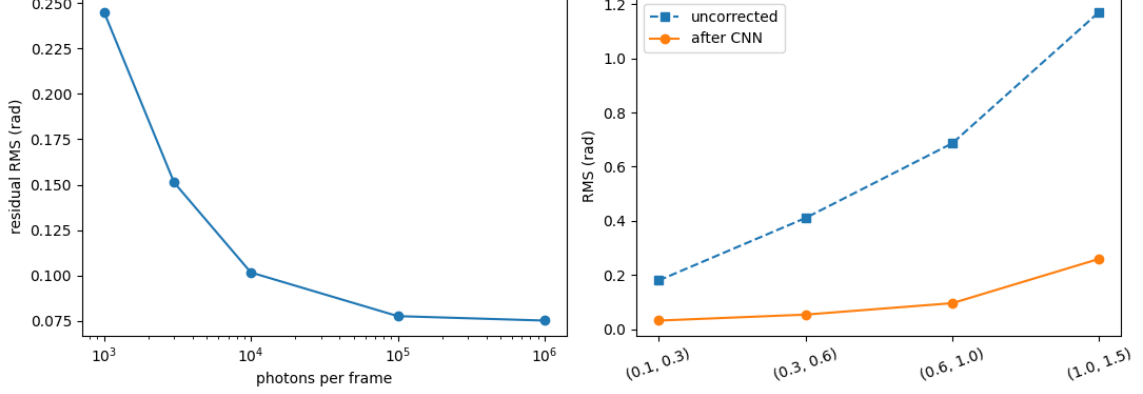


Figure 3: Residual RMS and Strehl ratio versus photon flux (left) and turbulence strength (right) for the CNN.

Table 4: Speed and accuracy on 200 frames (CPU). The physics fit uses the same renderer as the data (matched simulator).

Method	Residual RMS (rad)	Strehl	Time, 200 frames (s)	Iterative
Physics fit	0.024	0.999	37.9	Yes
CNN	0.074	0.993	0.711	No

on CPU, whereas the iterative fit took 37.9 s (about 5 frames per second). A single-pass estimator is attractive in principle for closed-loop AO systems, which typically run at high frame rates. These numbers, however, come from unoptimised research code on a CPU, exclude readout and control-loop latency, and were obtained without simulating temporal dynamics. They should be read as a comparative statement, not as evidence that either method meets the requirements of a specific AO system.

Effect of the matched simulator. The physics fit uses the same forward model that produced the data, so it is evaluated under idealised conditions (inverse crime [8]). The $3\times$ accuracy advantage in Table 4 could therefore be optimistic and might not survive realistic model mismatch, such as a different pupil geometry or turbulence generator. The CNN is also tested on the same generator it was trained on, so its performance under a different generator is untested. Neither method is favoured under mismatch in this study because mismatch was not tested.

Role of phase diversity. The ablation shows a larger residual for the one-channel CNN (0.214 vs. 0.077 rad), with the largest per-mode degradation for defocus and astigmatism. This matches the expected sign ambiguity of even-order modes in a single PSF. The size of the gap cannot be attributed solely to the missing channel, since the single-channel model was trained for fewer epochs.

Operating regime. At 10^3 photons per frame, the lowest flux in the trained range, the CNN reaches a mean Strehl ratio of 0.938, with larger residuals than at higher flux. In the turbulence sweep the residual grows with input strength, and the strong-turbulence result (0.260 rad) lies outside the training distribution and should be treated as an out-of-distribution observation. Both sweeps use one model and one seed.

Motivation for a hybrid. The CNN prediction is fast but the physics fit is considerably more accurate, so using the CNN prediction to initialise the physics fit is a natural combination.

Whether this reduces the number of iterations needed to reach the accuracy of the standard fit remains an open question.

8 Limitations and Future Work

- **Model mismatch (planned).** The physics fit uses the same simulator and may not withstand realistic mismatch, which may explain the lead reported in Table 4. Both methods should be tested on data from another generator (e.g., FFT-generated Kolmogorov phase screens or a pupil with a central obstruction), with matched and mismatched train/test pairs for the CNN.
- **Statistical confidence (planned).** Results come from one run of one seed. Future work should replicate the experiments with at least three seeds, reporting mean $\pm$ standard deviation, and also report the median and 90th percentile of the residual RMS.
- **Unequal-epoch ablation.** The single-channel model was trained for 7 epochs and the two-channel model for 15. The ablation should be repeated with equal training time.
- **Scale (planned).** The reduced profile (3000 training samples, 1.2 M-parameter CNN, 15 epochs) is expected to be a lower bound for the CNN. Final numbers should come from the full GPU profile.
- **Out-of-distribution coverage.** The photon sweep lies within the training range. The turbulence sweep has ranges partly below and completely above the training RMS range, so it mixes interpolation and extrapolation.
- **Hybrid estimator (planned).** The CNN-initialised fit has not been evaluated. Future work should report residual RMS and Strehl as a function of fit iterations for zero-start and CNN-initialised fits, the iterations needed to reach a fit tolerance, and the wall-clock time including the CNN forward pass. A negative result would also be informative.
- **Mode truncation.** Only 16 Zernike modes are modelled. Real turbulence has power in higher modes, which would appear as unmodelled residual.
- **Modal turbulence model.** The simulator uses a Kolmogorov-like modal spectrum, not full spatial phase screens.
- **Simulation only.** No laboratory or on-sky data were used, and no dynamic effects such as temporal lag or loop delay were modelled.
- **Timing.** Timings come from unoptimised code in one notebook session on CPU and are relative.
- **Subset difference.** The CNN residuals in Table 1 (full validation set) and Table 4 (first 200 validation samples) come from different subsets.

9 Conclusion

A small two-channel CNN successfully retrieved 16-mode wavefronts from phase-diverse PSFs, reducing the mean residual RMS from 0.549 to 0.077 rad. Removing the defocused channel raised the residual to 0.214 rad, as expected from the sign ambiguity of even-order modes, but the single-channel model was trained for fewer epochs and the comparison should be repeated at equal epochs. A gradient-descent physics fit was more accurate (0.024 rad) but about 53 times slower on CPU; it used the same renderer as the data, so its advantage may not hold under more realistic conditions. The photon-flux sweep (within the training range) and the turbulence sweep (interpolation plus partial extrapolation) indicate where the CNN remains effective. A hybrid estimator that initialises the fit from the CNN output is proposed but not yet tested. The conclusions hold only for this simulated environment, a single random seed and a limited training profile; real-system performance, and whether the physics fit’s advantage persists under model mismatch, remain to be shown.

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